SHREE SWAMINARAYAN COLLEGE OF NURSING

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SUBJECT: NURSING EDUCATION

DETAILED LESSON PLAN

TOPIC: VASA PREVIA

Submitted to:
Shree Swaminarayan College of Nursing, Kalol

ubmitted by:
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1st Year M.Sc. Nursing

Date: 26/10/2026

Lesson plan prepared for classroom teaching in Obstetrics and Gynaecological Nursing.

LESSON PLAN

Name of the student	Sathvara Priyanshi
Course / Year	1st Year M.Sc. Nursing
Subject	Nursing Education / Obstetrics and Gynaecological Nursing
Topic	Vasa Previa
Date	26/10/2026
Time	Approximately 35 minutes
Venue	Classroom
Method of teaching	Lecture-cum-discussion, questioning and explanation using a diagram/case scenario
Audio-visual aids	PowerPoint presentation, board/marker, diagram of placenta and fetal vessels, ultrasound/colour Doppler image (if available)

GENERAL OBJECTIVE

At the end of the teaching session, students will gain knowledge about vasa previa, including its meaning, definition, types, pathophysiology, risk factors, diagnosis, complications and nursing management.

SPECIFIC OBJECTIVES

- Define vasa previa and explain its clinical significance.
- Describe the types and basic pathophysiology of vasa previa.
- Identify the major risk factors and associated placental/cord abnormalities.
- Explain the signs, symptoms and diagnostic investigations.
- Discuss maternal-fetal complications and the principles of medical/obstetric management.
- Describe nursing responsibilities before, during and after delivery.
- State key discharge teaching and emergency warning signs.

TEACHING PLAN / PRESENTATION

The sequence below follows the tabular structure of the supplied lesson-plan sample: time, specific objective, content, teaching-learning activity, A.V. aids and evaluation.

SR. NO.	TIME	SPECIFIC OBJECTIVE	CONTENT	TEACHING–LEARNING ACTIVITY	A.V. AIDS	EVALUATION
1	1 min	**Introduction**	Good morning. I am Sathvara Priyanshi, a 1st Year M.Sc. Nursing student. Today's topic is **Vasa Previa**, an uncommon but serious obstetric condition in which fetal blood vessels lie close to or cross the internal cervical opening without the normal protection of placental tissue or Wharton's jelly.	Introduction and interaction	PPT / topic title	Students are oriented to the topic.

SR. NO.	TIME	SPECIFIC OBJECTIVE	CONTENT	TEACHING–LEARNING ACTIVITY	A.V. AIDS	EVALUATION
2	3 min	**Meaning and definition**	**Meaning:** “Vasa” refers to vessels and “previa” indicates that the vessels lie in front of/over the presenting part or cervical opening. **Definition:** Vasa previa is a condition in which unprotected fetal blood vessels run through the fetal membranes over or very near the internal cervical os. These vessels may rupture when the membranes rupture or during labour, causing rapid fetal blood loss. **Clinical importance:** The blood in these vessels belongs to the fetus, not the mother. Therefore, even a relatively small amount of bleeding can cause severe fetal anaemia, hypovolaemia or death.	Lecture-cum-discussion; explain key terms	PPT / labelled diagram	Define vasa previa and explain why it is dangerous.
3	3 min	**Types of vasa previa**	**Type I:** Associated with velamentous insertion of the umbilical cord. The vessels leave the cord and travel through the membranes without Wharton’s jelly before reaching the placenta; they may cross or lie close to the internal os. **Type II:** Associated with a succenturiate (accessory) placental lobe or bilobed placenta. Fetal vessels connecting the placental lobes pass through the membranes over or near the internal os. **Low-lying fetal vessels:** Some guidelines distinguish vessels close to, but not meeting the diagnostic threshold for, vasa previa. The exact distance criterion varies across guidance; clinical interpretation should follow local specialist protocols.	Explain and compare the two types	PPT / placenta diagrams	Differentiate Type I from Type II.
4	4 min	**Pathophysiology**	1. Abnormal cord insertion or an accessory/bilobed placenta develops. 2. Fetal vessels course through the membranes rather than being protected within the umbilical cord or placental tissue. 3. When these vessels pass over or close to the internal cervical os, they are exposed to pressure and stretching as the cervix dilates and the presenting part descends. 4. Spontaneous rupture of membranes or artificial rupture of membranes may tear the exposed vessels. 5. Fetal blood may be lost rapidly because the fetal circulating blood volume is small. 6. The fetus can develop acute anaemia, hypovolaemic shock, abnormal fetal heart rate, hypoxic injury or death unless delivery and neonatal resuscitation are immediate.	Stepwise explanation with flow chart	PPT / flow chart	Explain the sequence leading to fetal compromise.
5	3 min	**Risk factors**	• Velamentous insertion of the umbilical cord. • Succenturiate or bilobed placenta. • Low-lying placenta or placenta previa in the second trimester, even if the placental edge later moves away from the cervix. • Multiple pregnancy, especially when placental/cord anatomy is abnormal. • Pregnancy conceived with assisted reproductive technology/IVF has been associated with increased risk in studies. • Previous ultrasound finding of abnormal cord insertion or accessory placental lobe. **Important:** Vasa previa can occur without obvious risk factors; risk assessment does not replace ultrasound assessment when indicated.	Questioning and explanation	PPT / risk-factor list	Name at least four risk factors.

SR. NO.	TIME	SPECIFIC OBJECTIVE	CONTENT	TEACHING–LEARNING ACTIVITY	A.V. AIDS	EVALUATION
6	4 min	**Clinical features and complications**	**Before labour:** Often there are no symptoms; diagnosis is frequently made on ultrasound. **Warning signs in labour:** Vaginal bleeding after rupture of membranes together with sudden fetal heart-rate abnormality, bradycardia or a sinusoidal pattern should raise urgent concern. The bleeding may be fetal in origin. Do not delay emergency action while trying to prove the diagnosis if fetal compromise is present. **Fetal/neonatal complications:** Acute fetal haemorrhage, severe anaemia, hypovolaemia, hypoxia, need for urgent transfusion/resuscitation, neurological injury and perinatal death. **Maternal considerations:** The principal immediate danger is to the fetus; maternal bleeding may be limited, so the apparent amount of vaginal blood does not reliably indicate fetal blood loss.	Case-based discussion	PPT / case scenario	Identify warning signs requiring immediate escalation.
7	4 min	**Diagnosis and investigations**	**1. Ultrasound:** Routine mid-trimester fetal anatomy ultrasound may identify low placenta, abnormal cord insertion or accessory placental lobes. **2. Transvaginal ultrasound with colour Doppler:** Key test to map fetal vessels in relation to the internal cervical os; it is generally safe when performed by a trained clinician. **3. Pulsed-wave Doppler:** May help confirm that the waveform is fetal in origin when needed. **4. Follow-up scan:** Repeat assessment in the third trimester, often around/after 32 weeks according to local protocol, to confirm whether the condition persists and to guide birth planning. **5. Intrapartum suspicion:** Bleeding after membrane rupture with acute fetal compromise is an obstetric emergency. Clinical response must not be delayed for confirmatory testing. **Differential diagnosis:** Placenta previa, placental abruption, cervical/vaginal bleeding and other causes of fetal heart-rate abnormality.	Explain investigations and interpretation	PPT / Doppler image	Which investigation best maps vessels near the cervix?
8	4 min	**Medical and obstetric management**	**Antenatal planning:** Refer to an obstetrician/maternal-fetal medicine specialist. Confirm persistence with follow-up ultrasound, assess symptoms and preterm-birth risk, and plan delivery in a facility with emergency caesarean capability and neonatal resuscitation/transfusion support. **Admission:** Hospitalisation may be considered for selected patients, especially with bleeding, contractions, previous preterm birth, multiple pregnancy or long travel distance to hospital. The decision is individualised; routine admission for every asymptomatic patient is not universal practice. **Corticosteroids:** Consider antenatal corticosteroids when preterm delivery is likely, following gestational age and local guidelines. **Mode and timing of birth:** Planned caesarean delivery before labour or membrane rupture is generally recommended for confirmed persistent vasa previa. Guidance commonly discusses planned delivery in the late-preterm period (often around 34–36 or 34–37 weeks), but timing must be individualised by the specialist team to balance vessel rupture against prematurity. **Emergency:** If labour or rupture of membranes occurs with known vasa previa, or rupture is suspected with bleeding/fetal compromise, arrange immediate obstetric emergency response and urgent caesarean delivery as indicated. Prepare neonatal resuscitation and blood products if required.	Lecture-cum-discussion; emphasise emergency priorities	PPT / management algorithm	State the main principle of management.

SR. NO.	TIME	SPECIFIC OBJECTIVE	CONTENT	TEACHING–LEARNING ACTIVITY	A.V. AIDS	EVALUATION
9	5 min	**Nursing management**	**A. Antenatal care** • Review ultrasound reports and ensure the woman understands the diagnosis and follow-up plan. • Assess vital signs, bleeding, uterine activity, pain and fetal movements as indicated. • Explain the reason for specialist follow-up and planned caesarean delivery; provide emotional support and allow questions. • Teach the woman to seek urgent assessment for vaginal bleeding, contractions, fluid leakage, reduced fetal movements or abdominal pain. • Reinforce the individualised plan regarding activity, travel, admission and when/how to contact the maternity unit; do not prescribe blanket bed rest without an order. **B. Preparation for planned delivery** • Verify informed consent, identity, gestational age, allergies, investigations and preoperative checklist. • Ensure IV access, blood grouping/cross-match as ordered, and readiness of theatre, obstetric and neonatal teams. • Communicate clearly that fetal vessels must be protected from rupture; avoid procedures that could rupture membranes unless specifically directed by the obstetric team. • Confirm neonatal team availability and readiness for resuscitation and transfusion if required. **C. If bleeding, labour or membrane rupture occurs** • Call for immediate obstetric emergency assistance; document the time and circumstances. • Assess maternal ABCs, vital signs and bleeding while urgently monitoring fetal heart rate where feasible. • Do not delay escalation or delivery to obtain nonessential tests. • Prepare for emergency caesarean delivery; maintain IV access and administer prescribed treatment. • Communicate the suspected diagnosis and fetal status to the obstetrician, anaesthesia and neonatal teams. **D. After birth** • Support neonatal resuscitation and assessment for anaemia/hypovolaemia; facilitate blood sampling and transfusion if prescribed. • Monitor the mother after caesarean delivery, including vital signs, pain, wound, uterine tone and bleeding according to routine postoperative protocol. • Document events, interventions, fetal/neonatal condition and handover; provide debriefing and emotional support to the family.	Demonstration / discussion of nursing priorities	PPT / checklist	List nursing actions in an emergency.
10	2 min	**Prevention, counselling and health education**	There is no proven intervention that prevents the abnormal vessel arrangement once it has developed. The key preventive strategy is antenatal recognition, follow-up imaging and planned birth before labour or membrane rupture. Explain the diagnosis in clear, non-alarming language, provide written emergency contact instructions, and encourage attendance at all scheduled appointments. Counselling should include the reason for caesarean delivery, possible late-preterm birth, neonatal support and individualised decisions about admission.	Question-and-answer	PPT / patient education points	What advice should be given to the pregnant woman?
11	2 min	**Conclusion and evaluation**	Vasa previa is an uncommon but potentially life-threatening condition because unprotected fetal vessels lie over or near the cervical opening. Antenatal diagnosis with ultrasound and colour Doppler, specialist follow-up, planned caesarean delivery before labour/rupture of membranes, and rapid emergency response are central to improving fetal outcomes. Nurses play a key role in education, surveillance, preparation, escalation and family support.	Summarise and ask oral questions	PPT / board	Students summarise definition, diagnosis and nursing management.

ADDITIONAL TEACHING NOTES

1. Important points for clinical practice

• Vasa previa is a fetal-vessel condition; it is different from placenta previa, in which placental tissue covers or approaches the cervical os.

• A small volume of fetal blood loss can be catastrophic. Maternal vital signs may remain relatively stable despite severe fetal compromise.

• Colour Doppler transvaginal ultrasound is central to antenatal diagnosis and mapping the vessel location.

• A suspected rupture with fetal compromise is an emergency. Do not delay urgent delivery while trying to confirm the diagnosis.

• The timing of planned delivery and decisions about hospital admission must be individualised by the obstetric specialist and follow local guidelines.

2. Difference between placenta previa and vasa previa

Feature	Placenta previa	Vasa previa
Structure involved	Placental tissue lies over or near the internal cervical os.	Unprotected fetal blood vessels lie over or near the internal cervical os.
Source of bleeding	Usually maternal blood from placental tissue.	May be fetal blood from ruptured fetal vessels.
Main immediate danger	Maternal haemorrhage and related complications.	Rapid fetal blood loss, anaemia, hypovolaemia and fetal death.
Key diagnostic aid	Ultrasound assessment of placental location.	Ultrasound with colour Doppler to identify fetal vessels near the cervix.
Usual delivery planning	Depends on placental position, bleeding and clinical assessment.	Planned caesarean before labour/membrane rupture is generally recommended when persistent.

3. Evaluation questions

• Define vasa previa.

• Describe Type I and Type II vasa previa.

• Explain the pathophysiology of vasa previa in sequence.

• List four risk factors for vasa previa.

• Which ultrasound technique is particularly useful for mapping fetal vessels near the internal cervical os?

• Why is vaginal bleeding after rupture of membranes with acute fetal heart-rate abnormality an emergency?

• State the main principle of delivery planning in confirmed vasa previa.

• Mention five nursing responsibilities in the antenatal period and during an emergency.

4. Suggested teaching-learning flow chart

Abnormal cord insertion / accessory placental lobe → unprotected fetal vessels in membranes → vessels over/near internal cervical os → risk of vessel rupture during labour or membrane rupture → acute fetal haemorrhage and compromise → urgent obstetric response and delivery.

5. Conclusion

At the end of the session, students should understand that early recognition of vasa previa, appropriate follow-up imaging, a planned birth strategy and rapid escalation during bleeding or labour are essential. Nursing care focuses on education, assessment, coordination of the multidisciplinary team, emergency preparedness, neonatal support and accurate documentation.

6. Bibliography / references

1. Society for Maternal-Fetal Medicine (SMFM). Consult Series #37: Diagnosis and Management of Vasa Previa. Reaffirmed 2024. Available at: https://publications.smfm.org/publications/215-society-for-maternal-fetal-medicine-consult-series-37/

2. Royal College of Obstetricians and Gynaecologists (RCOG). Vasa Praevia: Diagnosis and Management (Green-top Guideline No. 27b). Available at:

https://www.rcog.org.uk/guidance/browse-all-guidance/green-top-guidelines/vasa-praevia-diagnosis-and-management-green-top-guideline-27b/

3. Jain V, et al. Guideline No. 439: Diagnosis and Management of Vasa Previa. Journal of Obstetrics and Gynaecology Canada, 2023. Available at: https://pubmed.ncbi.nlm.nih.gov/37209787/

Note: This is an educational lesson plan. Individual clinical decisions, including admission, corticosteroid use and timing of delivery, must be made by the responsible obstetric team according to the patient's condition and local protocols.